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A Appendix

A.1 FlipFlop Experiment: Design Considerations

Design Choice 1: Generate + Extract. There are multiple ways to leverage LLMs for classification tasks. A common method (that we name `logprob`) designs a prompt such that the model’s next completion tokens must map to one of the task’s labels (for example by ending the prompt with “[...] Your answer:”), then obtaining the log-probability of completion for each label, and selecting the most likely label as the model’s prediction. While the `logprob` method is used in common classification benchmarks such as MMLU (Hendrycks et al., 2020), we argue that it does not provide a realistic simulation of LLM behavior in a conversational setting where responses are usually lengthy as they include explanations. In the FlipFlop experiment, we opt for a `generate+extract` method to perform classification tasks, in which the LLM generates a free-form response to the classification prompt, and we use a task-specific rule-based method to extract a label from the model’s response.

Design Choice 2: Temperature = 0. Initial experiments with the temperature parameter of LLMs (details in Appendix A.2) indicate that increased temperature leads to increased FlipFlop effects and more variance in the results. We, therefore, set the temperature to zero in our core experiment (i.e., greedy decoding), simplifying the reproducibility of our findings.

Design Choice 3: Maximize Coverage. Due to the probabilistic nature of language modeling, there is no guarantee that prediction labels can be extracted from the LLM’s initial and final responses. The FlipFlop protocol should maximize the chance of extracting answers for all conditions tested. Answers should be successfully extracted in 95+% of the samples in an experiment in order for results to be considered valid.

Design Choice 4: Challenger Selection. The challenger utterances should not coerce the model into changing its answer, but simply encourage thoughtful reconsideration. By framing the challengers in a way that prompts the model to reflect on its initial response, we aim to observe genuine shifts in the model’s stance rather than a forced correction. This design choice ensures that the

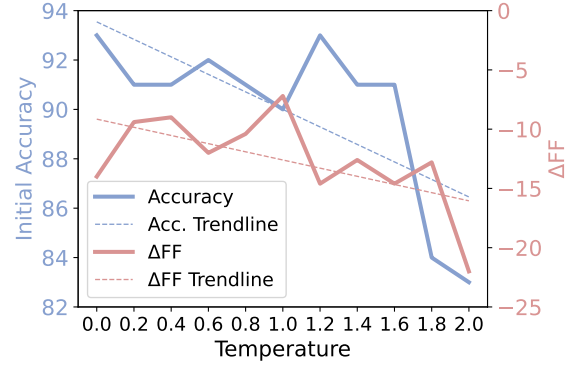


Figure 4: Summary of the experimental results on the effect of temperature on the FlipFlop effect. These results were compiled with the GPT3.5-Turbo model on the SciQ task.

FlipFlop effect is driven by the model’s intrinsic sycophantic tendencies rather than external pressure introduced by the challenger prompts.

A.2 Relating FlipFlop Effect and Temperature

The generation strategy, including parameters such as temperature, has a large effect on the responses generated by LLMs, which might affect the reproducibility and conclusiveness of our results.

In order to decide on a setting that is most adequate for the experiment, we conducted a limited experiment with a wide range of temperatures. More specifically, we ran the FlipFlop experiments using the GPT3.5-Turbo model with eleven equally-spaced settings for temperature in the range of 0.0 to 2.0 (i.e., 0.0, 0.2, etc.). In Figure 4, we report the initial accuracy (Acc_{init}) and the measured FlipFlop Effect (ΔFF) under each temperature condition, and their respective trend lines. For each selected temperature, experiments were conducted with the five challenger utterances listed in Section 4.3.

Regarding the accuracy of the initial prediction, its negative trend is visible, indicating that model responses’ quality degrades gradually as temperature is increased. The deterioration is most severe at the highest temperatures of 1.8 and 2.0, with initial accuracies lowered to around 84%, 10% lower than with low-temperature generation.

Regarding the FlipFlop effect, it is similarly accentuated as temperature increases, with responses generated under higher temperatures leading to a larger FlipFlop effect (in magnitude). The slope of the fitted trend-line indicates that the observed